

Stephen Huang

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EDUCATION

University of California, San Diego

September 2024 - Expected June 2028

B.S. in Electrical & Computer Engineering (*Expected June 2026*)

GPA: 3.85/4.00

M.S. in Electrical & Computer Engineering (*BS/MS Program; Expected June 2028*)

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Computer Architecture, Intro to ML

University of California, Irvine

September 2023 - June 2024

B.S. in Computer Science & Engineering

GPA: 3.82/4.00

TECHNICAL SKILLS

Languages: Python, HTML/CSS, Java, JavaScript/TypeScript, Verilog

Technologies: Git, GitHub Actions, Vue, Node.js, Express.js, React, PyTorch, PyBullet, Firebase, Google Cloud Functions

EXPERIENCE

UCI Engineering Student Council | *JavaScript, Express.js, Firebase, GitHub*

Irvine, CA

Technical Director

April 2024 - September 2024

- Designed a **Firebase (NoSQL)** data model and implemented serverless **REST APIs** (Node.js and Express.js on Google Cloud Functions) to create, update, and cancel bookings
- Integrated **Google OAuth** with backend **ID token verification** and role-based route protection to prevent unauthorized reservations

UCSD Existential Robotics Laboratory | *Python, Markdown, Bash*

San Diego, CA

Laboratory Assistant

October 2024 - Present

- Created a **Python** robotics simulation with **PyBullet** and **PyQt** for occupancy grid mapping and A* path planning as an interactive learning module for **50+ students**
- Developed **Bash** scripts to automate **Conda** environment setup, enabling new developers and non-programmers to run and modify the simulation smoothly
- Standardized tutorial website contributions through a **Markdown** style guide and reusable **HTML templates**, improving content consistency and reducing authoring overhead

PROJECTS

Distributed Social Network Client | *Python*

January 2024 - March 2024

- Developed a client for a course Distributed Social Network with a **CLI file manager** using pathlib, supporting safe operations, recursive listing, and profile management
- Integrated **OpenWeather** and **LastFM** APIs by parsing JSON and rendering user-configured content transclusions in posts
- Implemented a **TCP socket** client to the course server and a **Tkinter** interface, enabling real-time messaging with connection state handling

Spec to Verilog Converter | *Python, Verilog*

June 2025 - August 2025

- Built a **Python Verilog-generation tool** that maps I/O pin configurations into datapath routing from spec, using a **stack-based** algorithm to implement MUX chaining
- Generated a **4-module synthesizable Verilog** block per pin group across **10+ groups** (40+ total modules), including integration wrappers for mode control and **DFX/DFT hookup**
- Added **assertion-based validation** to check port declarations, internal connectivity, and mode-specific logic against the configured pin map, catching mismatches before integration

Space Typing Game | *TypeScript, Jest, Vite, GitHub*

September 2025 - December 2025

- Architected a **TypeScript** stage-progression pipeline that loads levels from JSON via a manifest, enabling stage additions and tuning without code changes
- Built a CI/CD pipeline in **GitHub Actions** with automated **Jest** testing, build validation on PRs, and automated deployment to GitHub Pages on PR merges
- Managed **sprint workflow** by scoping milestones and maintaining the **GitHub issue backlog**, keeping deliverables on schedule, and reducing coordination overhead